

Full Research Proposal

Project title

Multistatic 5G NR ISAC: Multi-Target Detection, and Tracking with Learning Based Receiver Selection

Project Aims

Integrated Sensing and Communication (ISAC) has emerged as a defining capability of 6G networks, reusing communication infrastructure and waveforms to sense the physical environment within existing spectral allocations. Within ISAC, the passive multistatic configuration is particularly attractive: distributed receivers exploit signals already transmitted for communication, using one transmitter and several spatially separated receivers to observe a target from multiple geometric perspectives. This diversity enables three-dimensional localization and improves robustness relative to a single monostatic sensor. The sensing geometry directly affects synchronization, observability, and estimation performance. In a monostatic configuration, transmitter and receiver are co-located, simplifying time alignment but requiring full-duplex operation and providing only a single aspect angle. In bistatic configuration, transmitter and receiver are spatially separated, removing the full-duplex requirement and expanding coverage, but introducing transmitter-to-receiver synchronization and bistatic geometry processing. A multistatic architecture combines multiple bistatic links, increasing geometric diversity and robustness to blockage and multipath-induced ambiguities.

This project's core novelty is a closed-loop sensing-tracking architecture: the multi-target tracker's predicted states — position, velocity, existence probability, and state uncertainty — together with the instantaneous multistatic sensing geometry, decides which receivers and antenna sectors should be used next. Measurements returned by the selected subset then update the tracker. Hence, sensing and tracking work together as a single feedback system, where each component influences and condition the other.

This is the central idea of the proposal and the basis for all the technical choices that follow.

The main objectives of the project are:

- Formulate receiver and antenna-sector activation as a constrained optimization problem driven by the tracker's predicted state, uncertainty, and existence probability, and learn a policy that approximates its solution under an energy, latency, and computation budget.
- Develop a labelled random-finite-set (RFS) multi-target tracker that preserves target identity under detection uncertainty, missed detections, and false alarms, with learning applied specifically to data association and motion-model adaptation.
- Close the loop between the two components and validate the resulting system in a physically grounded 5G NR simulation environment, quantifying the trade-off between sensing cost and tracking accuracy against established model-based and learning baselines.

Background

Recent work shows ISAC moving beyond monostatic toward multistatic sensing that is closer to real-world deployment. Reusing NR reference signals can support detection and localization, but performance depends heavily on sensing geometry, synchronization, receiver placement, and how realistic the channel model is. More receivers are not always better; they help only when they add new information, which is why receiver selection matters [6].

Another active area is multi-target sensing in realistic propagation conditions, where the challenge is no longer just detecting one target, but matching returns across receivers and keeping track of target identity over time in cluttered, moving scenes [2]. Detecting small, low-altitude objects such as uncrewed aerial vehicles (UAVs) is especially important. Drones can pose risks to airports, critical infrastructure, and public events, while conventional

Radar is often not ideal for small, slow, low-flying targets. By embedding sensing in the cellular network, the proposed approach can provide wide-area, always-on awareness without requiring dedicated radar hardware. Although UAVs are a motivating use case, the method is target-agnostic and can be applied wherever a distributed radio network needs to detect and track objects.

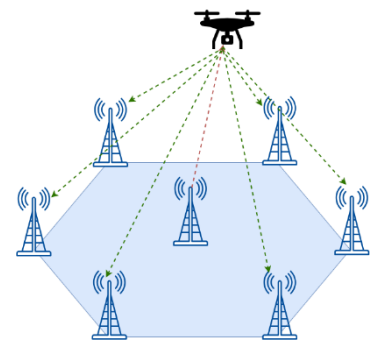


Figure 1: Topology

Two major practical challenges still limit multistatic ISAC. First, activating every receiver and antenna sector for every target is too expensive in terms of RF resources, sampling, and processing, so receiver choice must be made under uncertainty. Second, converting noisy detections into stable, identity-preserving tracks remains difficult in clutter, missed detections, and closely spaced targets. This project addresses both challenges together through the closed-loop framework described above.

Multistatic NR signal model

For transmitter Tx and receiver k observing a target at $u(t)$, the bistatic delay is

$$\tau_k(t) = \frac{R_{Tx}(t) + R_k(t)}{c} + \delta\tau_k,$$

where R_{Tx} and R_k are the Tx-target and target-Rx ranges, and $\delta\tau_k$ is a residual synchronization offset. The bistatic Doppler depends on how quickly the total path length changes, and the angle of arrival at receiver k is determined by its array geometry. Clutter is treated as a static background and suppressed before detection. Blockage is modelled with a per-link LoS/NLoS indicator, so the system only uses links that are actually available.

Range and velocity from the range-Doppler map

From the estimated PRS channel response $\hat{h}_{k,l,m}$, the delay profile is obtained using an M -point IDFT over the active subcarriers,

$$r(d) = \frac{1}{M} \sum_{k=0}^{M-1} \hat{h}_{k,l,m} e^{j2\pi kd/M},$$

whose peak at d_{peak} yields the bistatic range estimate

$$\hat{r} = \frac{c d_{\text{peak}}}{MK\Delta f},$$

where K is the PRS comb size, Δf is the subcarrier spacing, and c is the speed of light.

Similarly, an N_0 -point DFT across the PRS occasions estimates the Doppler response,

$$v(p) = \frac{1}{N_0} \sum_{m=0}^{N_0-1} \hat{h}_{k,l,m} e^{-j2\pi pm/N_0},$$

whose peak at p_{peak} provides the radial velocity estimate

$$\hat{v} = \frac{c}{f_c} \frac{p_{\text{peak}}}{N_0 P_{\text{PRS}}},$$

where P_{PRS} is the PRS periodicity and f_c is the carrier frequency. This estimation is performed independently at each receiver, and the resulting range and velocity measurements are used for multistatic fusion and target tracking.

Receiver selection as constrained optimization

At each epoch t , receiver selection is posed as

$$S_t^* = \arg \max_{S \subseteq \mathcal{K}} J_t(S)$$

subject to

$$|S| \leq N_{\text{max}}, \quad E_t(S) \leq E_{\text{max}}, \quad L_t(S) \leq L_{\text{max}},$$

where $S \subseteq \mathcal{K}$ is the selected receiver set, $J_t(S)$ is the utility function combining Fisher information, geometric observability, and target existence probability, and $E_t(S)$ and $L_t(S)$ are the corresponding energy and latency costs. Since solving this optimization exactly at every epoch is expensive, a learned policy is trained to approximate S_t^* in real time.

Tracking

Target tracking will be performed using the labelled multi-Bernoulli (LMB) filter, which jointly estimates target existence, state, and labels under clutter and missed detections. Learning is incorporated only for data association and adaptive motion-model selection, while the RFS filter remains the primary Bayesian estimator.

Significance & Innovation

The significance of this work is that it makes network embedded sensing more practical and efficient, helping advance an emerging 6G capability while also supporting safer situational awareness and the responsible use of autonomous systems in shared environments. Innovation has three main parts.

First, receiver selection is treated as a sequential decision problem rather than a one-time geometric rule. A reinforcement learning policy chooses receivers and antenna sectors, guided by model-based measures such as Fisher information and geometric dilution of precision, which indicate how informative the receiver geometry is for the predicted target location. This hybrid approach keeps the physical meaning of the model-based methods while adding the flexibility of learning, so the system can adapt to changing conditions and decide when a simple geometric choice is enough and when more receivers are needed.

Second, the tracking framework remains model-based at its core. It uses a labelled random finite set estimator to preserve target identity, while learning is applied only where it is most useful, such as motion-model adaptation. This keeps the design transparent and makes it easier to see where learning truly improves performance.

Third, both parts are combined in a single distributed pipeline and evaluated end to end, so the link between receiver selection and tracking performance is measured directly rather than assumed. The overall flow of the pipeline is shown in Figure 2.

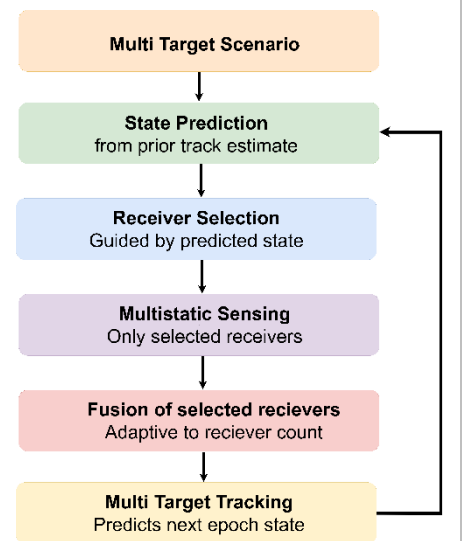


Figure 2: Flowchart

Proposed Research Plan, Methodology & Timeline

This project builds on an existing MATLAB framework for multistatic 5G NR sensing based on 3GPP TR 38.901. The current setup includes one transmitter and up to six receivers arranged in a hexagonal layout, together with an OFDM sensing chain for channel estimation, range - Doppler processing, CFAR detection, angle estimation, and cross receiver fusion.

A. Multi-bistatic sensing in 3D

The first part of the project focuses on 3D sensing with multiple Tx - Rx pair. Under TR 38.901, propagation is rich in multipath, often non-line-of-sight, and may involve multiple targets at once. For each Tx - Rx pair, the system estimates bistatic range and velocity from the range - Doppler map, while angle of arrival is estimated using standard array-processing methods such as beamforming, FFT beam space, Bartlett, or MUSIC. A single receiver is often not enough to recover an accurate 3D position, especially when geometry is poor. For that reason, the project uses multiple viewpoints, with the aim of combining at least three independent links for reliable localization and velocity estimation.

B. Learning-based, geometry-informed receiver selection

The second part of the project asks a practical question: which receivers should be activated at each time step? The idea is to first filter candidate receivers based on geometry and link quality and then let a reinforcement learning agent choose the subset that gives the best sensing value within a fixed budget.

The reward is guided by information-theoretic measures such as Fisher information, CRLB reduction, and observability. This makes the policy physically meaningful, rather than purely data driven. At the same time, a simple tiered selection strategy will be kept as an interpretable baseline for comparison.

C. Multi-receiver fusion

Once the useful measurements have been selected, they are fused using a weighted nonlinear least squares estimator. This method combines the selected range, Doppler, and angle measurements into a single target state estimate while accounting for the different quality of each link. Its main advantage is that it is simple, interpretable, and well suited to the multistatic geometry used in this project. Alternative methods such as Bayesian fusion, information filtering, covariance intersection, and learned fusion will be evaluated as comparison baselines. The main goal is robustness. Since the system intentionally works with only a subset of receivers, the fusion stage must still perform well when some measurements are weak, missing, or blocked.

D. Closed-loop multi-target tracking

The fused measurements are then passed to a multi-target tracker based on the LMB filter. This tracker maintains the target state, uncertainty, existence probability, and identity over time. Its output is then fed back to the receiver selection module, closing the loop so that future sensing decisions are based on the latest tracking information.



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Performance of the system will be measured not only in terms of localization accuracy, but also in terms of miss detections, identity switches, and sensing cost. This makes the trade-off between accuracy and resource use explicit.

E. Infrastructure-free hybrid sensing

In Year 3, the project will investigate a hybrid sensing architecture that combines multistatic NR with Rydberg-atom RF sensing to reduce dependence on dedicated radar hardware. The main objective is to determine whether a compact Rydberg receiver can provide useful complementary information, especially angle and high sensitivity field measurements in scenarios where conventional antenna geometry is limited or receiver activation must be reduced.

The approach is to integrate Rydberg measurements into the same labelled multi-target tracking framework used for NR. NR will continue to provide range and Doppler, while the Rydberg sensor will contribute additional field and angular information. A selection policy will be developed to decide when activating the Rydberg sensor improves tracking enough to justify its added energy, latency, and processing cost. The hybrid system will be evaluated against NR-only and radar-assisted baselines under matched simulation conditions, using localization error, GOSPA, identity switches, and resource usage as the main metrics.

The proposal timeline is shown below,

Year	Milestones
1	Build the geometry-based screening tools and model-based evaluation metrics, including Fisher information and geometric dilution of precision (GDOP). Set up the simulation environment, define the performance metrics, and train the reinforcement learning receiver selection policy, comparing it with greedy, random, and fixed baselines.
2	Develop the LMB tracker and evaluate its performance against CRLB-based limits. Add learning for data association and motion-model adaptation and then bring receiver selection and tracking together in one closed-loop system.
3	Test the full system across different target densities, heights, and motion patterns. Complete the Rydberg proof-of-concept.

Evaluation Plan

The evaluation will compare the proposed methods against strong baselines for both receiver selection and tracking. For receiver selection, these baselines include all-receiver activation, random selection, highest SNR selection, GDOP-based selection, greedy Fisher-information selection, and conventional reinforcement learning without geometric structure. For tracking, the methods will be compared against Extended Kalman Filter-Joint Probabilistic Data Association (EKF-JPDA), and an LMB tracker without learning. Performance will be measured using 3D localization RMSE, generalized optimal subpattern assignment (GOSPA), miss detections, identity switches, receiver activation ratio, latency, and energy use. Together, these metrics will show not only how accurate the system is, but also how efficiently it uses sensing resources.

Hypotheses and deliverables

The project is based on three main hypotheses. First, a closed-loop selection strategy guided by Fisher information will reduce the number of active receivers while maintaining good tracking performance. Second, learning-based data association inside the LMB tracker will reduce identity switches in cluttered, low-SNR environments. Third, hybrid Rydberg based 5G sensing will improve localization performance while reducing dependence on dedicated antenna array infrastructure.

This proposal is grounded in prior work on multistatic sensing and localisation. GLOBECOM 2025 results already show baseline multistatic 3D localisation/velocity performance as a function of active receiver count [6], motivating the receiver-selection problem addressed here, and this line of work is being extended in ongoing submissions to ISAC 2026 and ICC 2027. Together, these works demonstrate both the technical foundation and the research continuity needed for the proposed closed-loop sensing and tracking framework.

Anticipated Outcomes & Benefits

The project will deliver a validated closed-loop sensing and tracking system that reduces sensing cost while maintaining comparable accuracy, a labelled RFS tracker with carefully controlled learning components, and a reproducible evaluation framework suitable for comparison against strong baselines. The results will be



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disseminated through leading journals and conferences, and the work will be carried out in collaboration with industry partners to support deeper exploration of practical 6G sensing problems and contribute to ongoing 3GPP discussions on sensing for future networks. The project uses 3GPP TR 38.901 channel and waveform models. Reusing existing NR reference signals avoids the need for extra spectral allocation, but it still incurs computational, energy, and processing cost, which will be measured explicitly. Passive RF sensing is also less intrusive than camera-based surveillance because it does not collect identifiable imagery.

Alignment & Research Environment

This proposal fits naturally with Deakin's Building Safe and Secure Communities theme, particularly its focus on preventing threats and using emerging technologies responsibly. It also builds on an ongoing technical discussions with Prof. Shiva R. Pokhrel's research group at Deakin, whose recent work addresses complementary areas including uplink-reference-signal-based UAV sensing and 3D tracking on an end-to-end O-RAN testbed [9], data-aided UAV sensing frameworks [10], and federated learning for autonomous navigation and timing [11]. Reference [10] shows 3D observability is achievable for a single bistatic pair on live O-RAN infrastructure. This proposal tackles an even harder problem - closing the loop between prediction and sensing, using a learned policy to select which receivers and sectors to activate across a multistatic network, then fusing results within a labelled multi-target tracker that preserves identity under clutter and missed detections., which efficiently achieving observability for multiple simultaneous targets, rather than for one link at a time. In effect, these two lines of work tackle complementary halves of the same 3D multi-target ISAC sensing problem, and the applicant and host group are already in active discussion about how to bring them together as part of this fellowship [12].

Beyond the technical aspects, the motivation is simple. A dense cellular network that can detect, localize, and track objects in its surrounding airspace using communication signals it already transmits could provide a safer and more practical way to monitor shared environments without requiring dedicated radar infrastructure. Although UAVs are the main focus of this work, the approach is not limited to them. The same closed-loop sensing framework could be adapted to detect and track other types of objects as needed.

References

- [1] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," TR 38.901, 2020.
- [2] F. Liu et al., "Integrated Sensing and Communications: Toward Dual-Functional Wireless Networks for 6G and Beyond," IEEE JSAC, vol. 40, no. 6, 2022.
- [3] B.-N. Vo, B.-T. Vo, D. Phung, "Labeled RFS and the Bayes Multi-Target Tracking Filter," IEEE Trans. Signal Process., vol. 62, no. 24, 2014.
- [4] Z. Han et al., "Cellular Network Based Multistatic ISAC Systems," IET Commun., 2024.
- [5] A. K. Robinson, N. Prajapati, D. Senic, M. T. Simons, C. L. Holloway, "Determining the AoA of an RF Source with a Rydberg Atom-Based Sensor," Appl. Phys. Lett., vol. 118, 2021.
- [6] A. Sahoo, A. Sonny, S. Blandino, T. Ropitault, D. Griffith, T. Nguyen, N. Golmie, "Cooperative Multistatic Sensing Using 5G NR PRS Signal," IEEE GLOBECOM Workshop WS16, Taipei, 2025.
- [7] A. Sonny, Yeduri, S.R. and Cenkeramaddi, L.R., "Q-learning-based unmanned aerial vehicle path planning with dynamic obstacle avoidance," Applied Soft Computing, pp. 110773, 2023
- [8] A. Sonny and L. R. Cenkeramaddi, "Q-Learning-Based Multi-UAV Path Planning with Dynamic Obstacle Avoidance," 2026 18th International Conference on COMmunication Systems and NETworks (COMSNETS), Bengaluru, India, 2026, pp. 627-632.
- [9] Gurung, Arun K., Satha K. Sathananthan, and Shiva R. Pokhrel. "5G ISAC-Based UAV Detection and 3-D Tracking Using Uplink Sounding Reference Signals on an End-to-End O-RAN Simulation Testbed." arXiv preprint arXiv:2608.05826 (2026).
- [10] M. Nemati, S. R. Pokhrel and J. Choi, "Modelling Data Aided Sensing With UAVs for Efficient Data Collection," in IEEE Wireless Communications Letters, vol. 10, no. 9, pp. 1959-1963, Sept. 2021.
- [11] S. R. Pokhrel, "Quantum Position Navigation Timing for Autonomous Vehicles: A Federated Learning Framework," in IEEE Transactions on Vehicular Technology, vol. 75, no. 7, pp. 12697-12706, July 2026.
- [12] https://github.com/amala25/6G_MULTISTATIC_ISAC_WITH_TRACKING_DEAKIN_COLLAB